Machine Learning Course Project-I

Submission Deadline: 15th October 2026

Team Size: 3-4 members

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1. Project Overview

The course project will be conducted in the form of a **Kaggle competition** based on a supervised machine learning problem.

The objective of the project is to evaluate the complete machine learning approach followed by students, including data understanding, preprocessing, experimentation with different models, analysis of results, and final predictive performance.

The project will **not be evaluated solely on the basis of the Kaggle leaderboard**. Students are expected to demonstrate a systematic approach and a clear understanding of the work carried out.

2. Project Requirements

Students are expected to:

1. Perform appropriate **Exploratory Data Analysis (EDA)** and identify relevant observations about the dataset.
2. Apply and document the **data preprocessing** steps considered necessary for the problem.
3. Experiment with the **machine learning models taught in the course** up to the applicable stage of the project.
4. Maintain a record of the **models tried** and the **best score achieved** by each model.
5. Compare the performance of different models and discuss relevant observations from the experiments.
6. Identify and report the model that achieves the **best score on the public Kaggle leaderboard**.
7. Clearly document the overall approach, experiments, observations, and results in the final report.

Students are expected to make their own modelling and preprocessing decisions based on their understanding of the data and concepts covered in the course.

3. Model Experimentation

Experimentation with a variety of models forms an important component of the project.

Students should try the relevant models taught in the course and compare their performance. Obtaining a good score using a single model does not remove the requirement to experiment with other relevant models.

For every model attempted, the final report must include:

- Model name.
- Best score achieved.
- Key observations or learnings from that experiment.

A consolidated table similar to the following should be included in the report:

Model	**Best Score Achieved**
Model 1	–
Model 2	–
Model 3	–
Model 4	–
...	...

4. Final Report

The report should provide a concise but complete documentation of the work carried out during the project.

It should include the following:

1. **Problem and Dataset:** A brief description of the problem and dataset.
2. **EDA:** Important observations obtained from exploratory analysis.
3. **Preprocessing:** The preprocessing steps performed and relevant justification.
4. **Models Tried:** A record of all relevant models experimented with and the best score achieved by each.
5. **Results and Discussion:** A comparison of the experimental results and relevant observations.
6. **Best Model:** The model achieving the best public Kaggle leaderboard score.

The emphasis should be on clearly communicating the approach followed and the reasoning behind important decisions rather than unnecessarily documenting every intermediate experiment.

5. Kaggle Evaluation Guidelines

Students may use any of the models or approaches explored during the project for Kaggle submissions.

The report must clearly identify the model that achieved the student's best score on the public leaderboard.

The public leaderboard score will not be used for the final evaluation. Instead, we'll use the **scores calculated on the private test data to evaluate the teams and for the Kaggle leaderboard grading component**. Students should therefore focus on the complete project rather than optimizing exclusively for the leaderboard.

6. Evaluation Rubric

The project will be evaluated using the following rubric:

Component	**Weightage**
Data Preprocessing	2M
Variety of Models Tried	3M
Viva + Report	5M
Performance (Leaderboard)	5M
Total	**15M**

7. Important Notes

- Scores are calculated on the **private test data** to evaluate the teams for the Kaggle leaderboard grading component.
- The project is evaluated on the **overall machine learning approach**, not only the final Kaggle score.
- Students are expected to experiment with a **variety of relevant models** taught in the course.
- The **best score achieved by every model tried** must be recorded in the report.
- The model with the student's **best public leaderboard score** must be included in the report.
- Students should be able to explain and justify their work during the viva.

Note: The purpose of the project is to assess the student's ability to independently apply the machine learning concepts covered in the course to a supervised learning problem.